Carleton UNIVERSITY

**WINTER TERM MIDTERM I**
**EXAMINATION**
**April 16, 2025**

**DURATION: _3_ HOURS** **No. Of Students: 22**

**Department Name & Course Number:** **Systems and Computer Engineering**
**SYSC 5108**

**Course Instructor (s): Changcheng Huang**

AUTHORIZED MEMORANDA

*Nonprogrammable calculator*

**Students MUST count the number of pages in this examination question paper before beginning to write, and report any discrepancy to a proctor. This question paper has _____2_____pages + cover page = _3__ pages in all.**

**This examination question paper may be taken from the examination room.**

**In addition to this question paper, students require: an examination booklet yes**
**Scantron Sheet no**

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**Name:** ____________________________________________________________

**Student Number:** ____________________________________________________

1. (15%) Given the following dataset. Use the third sample point as the test data and the remaining as training data.

| $x$ | 0 | 1 | 3 | 6 | 9 |
|---|---|---|---|---|---|
| $y$ | -1 | 2 | 1 | 4 | 2 |

Find the optimal linear regression model manually. Find the mean squared error for the training data and the mean squared error for test data.

2. (15%) Consider the requirement to generalize logistic regression from two classes to $K$ classes, i.e., the output $y \in \{1, 2, \cdots, K\}$. Assume $y$ follow multinoulli distribution and the logistic regression model can be defined as

$$p(y = k|\boldsymbol{x}, \beta) = \mathrm{softmax}(z)_k = \frac{\exp(z_k)}{\sum_l \exp(z_l)}$$

where $z_k = \beta_{k0} + \beta_k^T x, k = 1, \cdots, K-1$ and $z_K = 0$. Then the MLE cost function can be written as

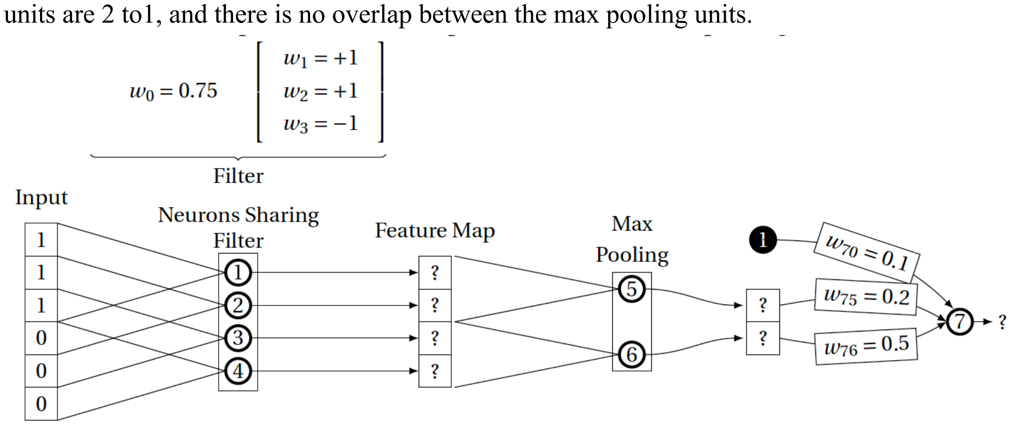
$$J(\beta) = -\sum_{i=1}^{M}\sum_{k=1}^{K} I_k(y_i) \log \mathrm{softmax}(z^{(i)})_k$$

Where $I_k(y_i) = 1, \text{if } y_i = k,$ and $0, \text{otherwise}$. Show that

$$\nabla_{\beta_k} J(\beta_k) = -\sum_{i=1}^{M}\sum_{k=1}^{K} I_k(y_i)(1 - \mathrm{softmax}(z^{(i)})_k)\boldsymbol{x}_i$$

Hint: $(\frac{u}{v})' = \frac{u'v - v'u}{v^2}$

3. (20%) The figure below illustrates a layer of a convolutional neural network that is processing a one-dimensional input. For ease of reference each of the neurons in the network has been labeled: 1; 2; 3; 4; 5; 6; 7. The architecture of the network consists of ReLUs that share a kernel (Neurons 1; 2; 3; 4), followed by a sub-sampling layer containing two max pooling units (Neurons 5; 6), and then a fully connected layer containing a single ReLU (Neuron 7). The kernel in the first layer has a window size 3, a bias, and a stride of 1. The max pooling units are 2 to1, and there is no overlap between the max pooling units.




(a) Assuming the target output for this input is 2, calculate the $\delta$ for each neuron in the network.
(b) Calculate the weight update for $w_{75}$ and $w_1$

4. (10%) Suppose $\gamma$= 0.9 and the reward sequence is $R_1$= 2 followed by an infinite sequence of 7s. What are $G_0$ and $G_1$?

5. (20%) Consider a two-state-two-action reinforcement learning scenario with the following environment.

| State | Action | Next State | Probability | Reward |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | -1 |
| 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 0 | 1 | -1 |

Assume $\gamma = 0.1$ and an $\epsilon$-greedy policy with $\pi(1|0) = \pi(0|1) = 0.6$.

(a) Starting from state 0 and 1 at time 0 respectively, generate two trajectories using the following random number sequence and terminate at $t = 3$. Assume $\alpha = 0.6$, estimate the state-value function and action-value function with Monte-Carlo first-visit method using the generated trajectories. Indicate clearly which random numbers you have used.

0.94 0.27 0.30 0.18 0.75 0.35

(b) Assume $\alpha = 0.6$, starting with all action values equal to 0, find action values using off-policy TD control with the trajectories generated in (a)

6. (20%) Stormy nights are rare. Burglary is also rare, and if it is a stormy night, burglars are likely to stay at home (burglars don't like going out in storms). Cats don't like storms either, and if there is a storm, they like to go inside. The alarm on your house is designed to be triggered if a burglar breaks into your house, but sometimes it can be set off by your cat coming into the house, and sometimes it might not be triggered even if a burglar breaks in (it could be faulty or the burglar might be very good).

(a) Model the relationship between stormy night or not, burglars at your house or not, cats at your house or not, alarm triggered or not as a directed model and show the directed graph.

(b) Using the following examples to estimate the probabilities that characterize the directed graph. Show the probabilities in the graph.

| ID | STORM | BURGLAR | CAT | ALARM |
|---|---|---|---|---|
| 1 | false | false | false | false |
| 2 | false | false | false | false |
| 3 | false | false | false | false |
| 4 | false | false | false | false |
| 5 | false | false | false | true |
| 6 | false | false | true | false |
| 7 | false | true | false | false |
| 8 | false | true | false | true |
| 9 | false | true | true | true |
| 10 | true | false | true | true |
| 11 | true | false | true | false |
| 12 | true | false | true | false |
| 13 | true | true | false | true |

(c) What value will the Bayesian network predict for ALARM, given that there is both a burglar and a cat in the house but there is no storm?

(d) Generate a sample that includes a random outcome for each random variable in the network by following the convention that a positive outcome is the state 1 while the negative is state 0 (e.g., for random variable STORM: $\text{true} = 1, \text{false} = 0$) with the following random number sequence. Indicate clearly which random numbers you have used.

0.94 0.27 0.30 0.18